Kalman Filter Forecast
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The *Kalman* filter (*Kalman*, 1960) is a recursive, optimal linear estimator designed to infer the hidden state of a dynamic system from noisy observations. Originally developed for aerospace engineering—most famously applied in the Apollo guidance computer—it has become one of the most consequential algorithms in statistics, engineering, and econometrics.

Classical time series models such as ARIMA treat the observed series directly. The state space framework subsumes ARIMA as a special case and, crucially, provides:

- A principled treatment of missing observations.
- A natural mechanism for time-varying parameters.
- The ability to decompose a series into interpretable latent components (trend, seasonal, cycle, irregular).
- A unified likelihood via the prediction error decomposition, enabling maximum likelihood estimation via the EM algorithm or direct numerical optimisation.

State Space Representation

Let $\boldsymbol{\alpha}_t \in \mathbb{R}^m$ denote the state vector at time t. This is an unobservable (latent) vector that encapsulates all information about the system needed to predict its future evolution. Examples include the local level, slope, or seasonal harmonic coefficients of a time series. Let $\mathbf{y}_t \in \mathbb{R}^p$ denote the observation vector at time t. This is what we actually measure, typically a noisy function of the state.

Linear Gaussian State Space Model
The general linear Gaussian state space model (SSM) consists of two equations:
Observation (Measurement) Equation:

$$\mathbf{y}_t = \mathbf{Z}_t \boldsymbol{\alpha}_t + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{H}_t)$$

State (Transition) Equation:

$$\boldsymbol{\alpha}_{t+1} = \mathbf{T}_t \boldsymbol{\alpha}_t + \mathbf{R}_t \boldsymbol{\eta}_t, \qquad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}_t)$$

where the system matrices are defined as follows:

Symbol	Dimension	Name	Role
$\mathbf{Z}_t$	$p \times m$	Design matrix	Maps state to observations
$\mathbf{T}_t$	$m \times m$	Transition matrix	Governs state dynamics
$\mathbf{R}_t$	$m \times r$	Selection matrix	Maps noise to state
$\mathbf{H}_t$	$p \times p$	Observation noise covariance	Measurement error variance
$\mathbf{Q}_t$	$r \times r$	State noise covariance	Process (evolution) noise variance

Initial Conditions: The state at $t = 1$ is initialized with prior distribution:
$\boldsymbol{\alpha}_1 \sim \mathcal{N}(\mathbf{a}_1, \mathbf{P}_1)$
where $\mathbf{a}_1$ is the prior mean and $\mathbf{P}_1$ is the prior covariance.

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Notation clarification: The subscript t on the system matrices allows them to be time-varying, enabling models with structural breaks or time-varying coefficients. In stationary or homogeneous models the matrices are constant and the subscript is dropped.

Fundamental Assumptions

The *Kalman* filter is optimal (in the sense of minimum mean squared error) when the following assumptions hold:

Assumption 1: Linearity
Both the observation and state transition equations are linear functions of the state vector. This is expressed in equations above. Non-linear systems require extended or unscented variants.

Assumption 2: Gaussianity
The noise processes ε_t and $\boldsymbol{\eta}_t$ are *Gaussian* (normally distributed). Under linearity and Gaussianity, all conditional distributions of the state given past observations remain *Gaussian*, making the mean and variance sufficient statistics for the full conditional distribution. Formally:

$$p(\boldsymbol{\alpha}_t \mid \mathbf{y}_{1:t}) = \mathcal{N}(\mathbf{a}_{t|t}, \mathbf{P}_{t|t})$$

Assumption 3: Mutual Independence of Noise Terms
The disturbance sequences \{\boldsymbol{\varepsilon}_t\} and \{\boldsymbol{\eta}_t\} are mutually independent at all lags:

$$\mathrm{Cov}(\boldsymbol{\varepsilon}_t, \boldsymbol{\eta}_s) = \mathbf{0} \quad \forall\ t, s$$

and each process is serially uncorrelated (white noise):

$$\mathrm{Cov}(\boldsymbol{\varepsilon}_t, \boldsymbol{\varepsilon}_s) = \mathbf{0}, \quad \mathrm{Cov}(\boldsymbol{\eta}_t, \boldsymbol{\eta}_s) = \mathbf{0} \quad \text{for } t \neq s$$

Assumption 4: Known System Matrices
The system matrices $\mathbf{Z}_t, \mathbf{T}_t, \mathbf{R}_t, \mathbf{H}_t, \mathbf{Q}_t$ are assumed known (or have been estimated from data before filtering begins). In practice, unknown hyperparameters are estimated by maximizing the prediction error decomposition likelihood (see Section 8).

Assumption 5: Known Initial Distribution
The initial prior $(\mathbf{a}_1, \mathbf{P}_1)$ is assumed known. Diffuse (non-informative) initialization is used when the initial state is unknown; this is handled via the diffuse *Kalman* filter of *Koopman & Durbin* (2003).

Mathematical Derivation

Bayesian Interpretation
The *Kalman* filter is a recursive application of *Bayes*' theorem to the state space model. Define:

- Predicted distribution (prior for t given data up to t-1): $p(\boldsymbol{\alpha}_t \mid \mathbf{y}_{1:t-1})$
- Filtered distribution (posterior for t given data up to t): $p(\boldsymbol{\alpha}_t \mid \mathbf{y}_{1:t})$

By the *Markov* property of the state equation, the model defines a hidden Markov model (HMM). The filter exploits this structure to propagate the sufficient statistics $(\mathbf{a}_{t|t}, \mathbf{P}_{t|t})$ forward in time.

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We adopt the notation of *Durbin & Koopman* (2012):

- $\mathbf{a}_{t|t-1} = \mathbb{E}[\boldsymbol{\alpha}_t \mid \mathbf{y}_{1:t-1}]$: one-step-ahead predicted state mean
- $\mathbf{P}_{t|t-1} = \mathrm{Var}[\boldsymbol{\alpha}_t \mid \mathbf{y}_{1:t-1}]$: one-step-ahead predicted state covariance
- $\mathbf{a}_{t|t} = \mathbb{E}[\boldsymbol{\alpha}_t \mid \mathbf{y}_{1:t}]$: filtered state mean
- $\mathbf{P}_{t|t} = \mathrm{Var}[\boldsymbol{\alpha}_t \mid \mathbf{y}_{1:t}]$: filtered state covariance

Derivation of the Kalman Gain

The prediction error (also called the innovation) at time t is:

$$\mathbf{v}_t = \mathbf{y}_t - \mathbf{Z}_t \mathbf{a}_{t|t-1}$$

This is the discrepancy between the new observation and what was predicted. The innovation covariance is:

$$\mathbf{F}_t = \mathbf{Z}_t \mathbf{P}_{t|t-1} \mathbf{Z}_t^\top + \mathbf{H}_t$$

The *Kalman* gain $\mathbf{K}_t \in \mathbb{R}^{m \times p}$ determines how much the predicted state should be revised in light of the innovation:

$$\mathbf{K}_t = \mathbf{P}_{t|t-1} \mathbf{Z}_t^\top \mathbf{F}_t^{-1}$$

Intuition for the *Kalman* gain: $\mathbf{K}_t$ balances the reliability of the model prediction against the reliability of the new observation. When $\mathbf{H}_t \to \mathbf{0}$ (no measurement error), $\mathbf{K}_t \to \mathbf{Z}_t^{-1}$ and the state is updated entirely by the observation. When $\mathbf{P}_{t|t-1} \to \mathbf{0}$ (very certain prior), $\mathbf{K}_t \to \mathbf{0}$ and the observation is ignored.

Derivation from the Conditional Normal Distribution
Because all random variables are jointly *Gaussian*, the conditional distribution of $\boldsymbol{\alpha}_t \; given \; \mathbf{y}_{1:t}$ can be derived using the conditional normal formula. For jointly normal random vectors:

$$\begin{pmatrix} \mathbf{x} \\ \mathbf{y} \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \boldsymbol{\mu}_x \\ \boldsymbol{\mu}_y \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{xx} & \boldsymbol{\Sigma}_{xy} \\ \boldsymbol{\Sigma}_{yx} & \boldsymbol{\Sigma}_{yy} \end{pmatrix} \right)$$

the conditional distribution is:

$$\mathbf{x} \mid \mathbf{y} \sim \mathcal{N}\big(\boldsymbol{\mu}_x + \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}(\mathbf{y} - \boldsymbol{\mu}_y),\ \boldsymbol{\Sigma}_{xx} - \boldsymbol{\Sigma}_{xy}\boldsymbol{\Sigma}_{yy}^{-1}\boldsymbol{\Sigma}_{yx}\big)$$

Applying this to $(\boldsymbol{\alpha}_t, \mathbf{y}_t)$ given $\mathbf{y}_{1:t-1}$ (both Gaussian with known means and covariances from the prediction step) yields exactly the *Kalman* update equations.

<u>Prediction–Update Cycle</u>

The *Kalman* filter iterates between two steps for $t = 1, 2, \ldots, T$:

Prediction Step (Time Update)
Project the state forward using the dynamics:

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$$\mathbf{a}_{t|t-1} = \mathbf{T}_{t-1}\,\mathbf{a}_{t-1|t-1}$$
$$\mathbf{P}_{t|t-1} = \mathbf{T}_{t-1}\,\mathbf{P}_{t-1|t-1}\,\mathbf{T}_{t-1}^{\top} + \mathbf{R}_{t-1}\,\mathbf{Q}_{t-1}\,\mathbf{R}_{t-1}^{\top}$$

Interpretation: the first equation propagates the mean forward; the second equation propagates the covariance forward, adding process noise that accounts for the stochastic evolution of the state.

Update Step (Measurement Update)
Upon receiving new observation $\mathbf{y}_t$, compute:

Innovation:
$$\mathbf{v}_t = \mathbf{y}_t - \mathbf{Z}_t\,\mathbf{a}_{t|t-1}$$

Innovation covariance:
$$\mathbf{F}_t = \mathbf{Z}_t\,\mathbf{P}_{t|t-1}\,\mathbf{Z}_t^{\top} + \mathbf{H}_t$$

Kalman gain:
$$\mathbf{K}_t = \mathbf{P}_{t|t-1}\,\mathbf{Z}_t^{\top}\,\mathbf{F}_t^{-1}$$

Filtered state mean:
$$\mathbf{a}_{t|t} = \mathbf{a}_{t|t-1} + \mathbf{K}_t\,\mathbf{v}_t$$

Filtered state covariance (Joseph form for numerical stability):
$$\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{Z}_t)\,\mathbf{P}_{t|t-1}\,(\mathbf{I} - \mathbf{K}_t\mathbf{Z}_t)^{\top} + \mathbf{K}_t\mathbf{H}_t\mathbf{K}_t^{\top}$$

The Joseph form of equation above is numerically preferred over the standard form $\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{Z}_t)\mathbf{P}_{t|t-1}$ because it guarantees that the covariance remains symmetric and positive semi-definite even when floating-point errors accumulate.

Forecasting Beyond the Sample
To forecast h steps ahead (for $h \geq 1$) from time T, we simply iterate the prediction step without an update:

$$\mathbf{a}_{T+h|T} = \mathbf{T}^{h}\,\mathbf{a}_{T|T}$$
$$\mathbf{P}_{T+h|T} = \mathbf{T}^{h-1}\left(\mathbf{T}\,\mathbf{P}_{T|T}\,\mathbf{T}^{\top} + \mathbf{R}\mathbf{Q}\mathbf{R}^{\top}\right)(\mathbf{T}^{h-1})^{\top} + \cdots$$

The h-step-ahead forecast of the observation is:

$$\hat{\mathbf{y}}_{T+h|T} = \mathbf{Z}\,\mathbf{a}_{T+h|T}$$

with associated forecast error covariance:

$$\mathbf{F}_{T+h|T} = \mathbf{Z}\,\mathbf{P}_{T+h|T}\,\mathbf{Z}^{\top} + \mathbf{H}$$
A $(1-\alpha)$-level prediction interval for a scalar observation is then:

$$\hat{y}_{T+h|T} \pm z_{\alpha/2}\sqrt{F_{T+h|T}}$$

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The Kalman Smoother
The Kalman filter produces filtered estimates $\mathbf{a}_{t|t}$ using only past and present data. The Kalman smoother (specifically the Rauch–Tung–Striebel smoother) uses the full dataset to produce smoothed estimates $\mathbf{a}_{t|T}$ for all $t \leq T$, which have lower variance:

$$\mathbf{L}_t = \mathbf{P}_{t|t}\,\mathbf{T}_t^\top\,\mathbf{P}_{t+1|t}^{-1}$$
$$\mathbf{a}_{t|T} = \mathbf{a}_{t|t} + \mathbf{L}_t\left(\mathbf{a}_{t+1|T} - \mathbf{a}_{t+1|t}\right)$$
$$\mathbf{P}_{t|T} = \mathbf{P}_{t|t} + \mathbf{L}_t\left(\mathbf{P}_{t+1|T} - \mathbf{P}_{t+1|t}\right)\mathbf{L}_t^\top$$

The backward recursion runs from $t = T - 1$ down to $t = 1$, initialized with $\mathbf{a}_{T|T}$ and $\mathbf{P}_{T|T}$ from the filter.

<u>Extensions: Non-Linear and Non-Gaussian Variants</u>

When assumptions 1 or 2 fail, the Kalman filter must be extended:

Extended Kalman Filter (EKF)
For non-linear state or observation equations:

$$\mathbf{y}_t = h(\boldsymbol{\alpha}_t) + \boldsymbol{\varepsilon}_t, \qquad \boldsymbol{\alpha}_{t+1} = f(\boldsymbol{\alpha}_t) + \boldsymbol{\eta}_t$$

the EKF linearizes f and h around the current state estimate using a first-order *Taylor* expansion:

$$\mathbf{T}_t \approx \left.\frac{\partial f}{\partial \boldsymbol{\alpha}}\right|_{\boldsymbol{\alpha}=\mathbf{a}_{t|t}}, \qquad \mathbf{Z}_t \approx \left.\frac{\partial h}{\partial \boldsymbol{\alpha}}\right|_{\boldsymbol{\alpha}=\mathbf{a}_{t|t-1}}$$

The *Jacobian* matrices $\mathbf{T}_t and \mathbf{Z}_t$ replace the constant system matrices. The EKF is a first-order approximation and can diverge when the non-linearity is severe.

Unscented Kalman Filter (UKF)
The Unscented Kalman Filter uses a deterministic sampling strategy—sigma points—to propagate the mean and covariance through non-linear functions without linearization. It achieves accuracy to at least the third order for any smooth non-linearity (*vs*. first order for the EKF). The sigma points $\{\boldsymbol{\mathcal{X}}_i\}$ are chosen as:

$$\boldsymbol{\mathcal{X}}_0 = \mathbf{a}_{t|t}, \quad \boldsymbol{\mathcal{X}}_i = \mathbf{a}_{t|t} \pm \sqrt{(m+\lambda)\mathbf{P}_{t|t}}_i, \quad i = 1, \ldots, m$$

where $\lambda = \alpha^2(m+\kappa) - m$ is a scaling parameter.

Particle Filters
For highly non-linear or non-*Gaussian* models, Sequential *Monte Carlo* (particle filters) represent the posterior distribution $p(\boldsymbol{\alpha}_t \mid \mathbf{y}_{1:t})$ by a set of N weighted random samples (particles) $\{(\boldsymbol{\alpha}_t^{(i)}, w_t^{(i)})\}_{i=1}^N$. This is fully non-parametric and converges to the true posterior as $N \to \infty$.

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